

Computer Vision For Volleyball Serve Analytics

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Introduction

This project develops a single-camera system for automated volleyball serve analysis. The tool processes standard baseline video and extracts objective measurements of a serve's flight path, landing location, and overall consistency.

Traditional volleyball serve review is performed manually: players scrub through video, analyze the trajectory, and visually estimate landing spots. This approach is slow, imprecise, and provides no quantitative feedback. By contrast, this system offers a streamlined workflow that produces precise, repeatable metrics from ordinary practice footage.

This system includes ball detection, temporal tracking, trajectory reconstruction, hit and landing-frame estimation, and court-mapped landing coordinates. Together, these components form an accessible analysis pipeline for players and coaches.

About Serves

The serve is the only action in volleyball performed entirely under one player's control and can directly lead to points. Even small variations in toss height, foot placement, contact angle, or timing can shift the ball's landing location by several meters. Precise, objective feedback on where serves land and how consistently they reach the intended target is essential for meaningful practice and long-term improvement.

Data Collection

Data was collected using a smartphone camera positioned at the center of the baseline, approximately 4 meters behind the court and 74 inches high. All sessions were recorded at 1080p, 60 FPS, ensuring the entire court was in frame.

Annotations include:

- Six court-corner points per session
- Ball bounding boxes per frame

The dataset consists of:

- 5 recordings
- 5 ball types
- 113 serves
- 16,172 frames



Pipeline

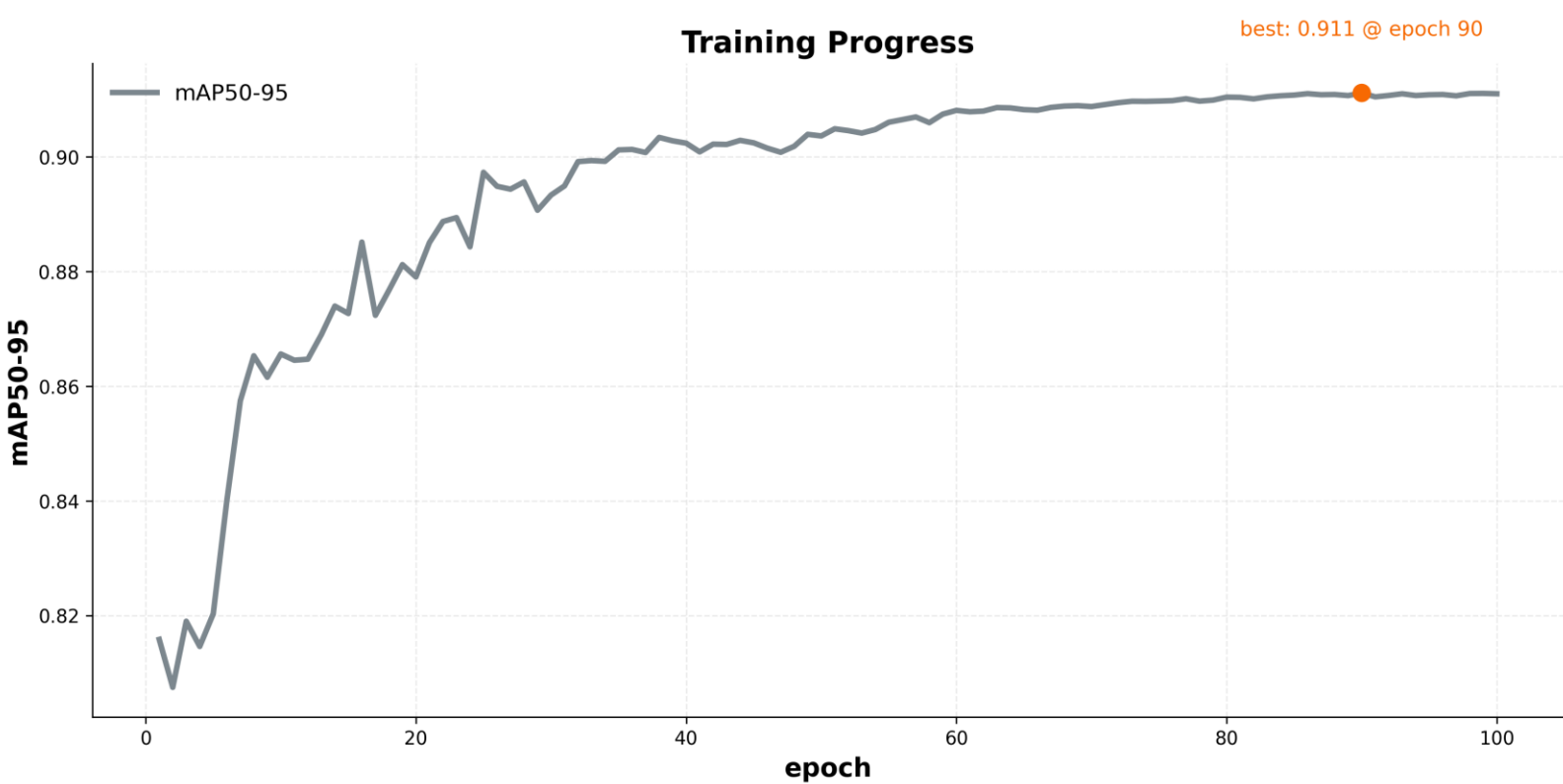
- 1) Serve Isolation:
Raw session footage is split into individual serve clips based on hand movement.
- 2) Ball Detection (YOLOv8):
A custom-trained YOLOv8 model detects the ball in every frame.
- 3) Ball Tracking (SORT):
SORT links detections across time and produces short trajectory fragments.
- 4) Trajectory Merging:
Fragments are combined using a greedy algorithm that minimizes gaps.
- 5) Final Trajectory Selection:
The merged chain with stable motion and realistic size decay is chosen as the serve.
- 6) Hit and Landing Frame Estimation:
Ball motion and size are smoothed with a Savitzky–Golay filter to estimate the hit and landing frames.
- 7) Court Homography:
Using six user-labeled court corners, pixel coordinates are projected into real space..
- 8) Trajectory & Landing Output:
The system generates plots, heatmaps, and serve-level statistics for interpretation.

Evaluation

YOLOv8 Performance

Trained for 100 epochs on 1920x1080 images:

- F1: 99.489%
- Precision: 99.452%
- Recall: 99.527%



Landing Accuracy

Landing-frame error is typically 2–3 frames, but this has negligible effect on landing-zone estimation because the ball's motion is almost stationary in the final frames near landing.

Conclusion

This project demonstrates that accurate volleyball serve analysis can be achieved using a single baseline camera. YOLOv8 and SORT provide reliable ball tracking, while homography enables real-court trajectory and landing estimation. The system offers accessible, objective data that helps players evaluate accuracy, consistency, and improvement over time.